

Math 161

Problem set 7 solutions

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1. (a) The statement “ $\emptyset$ is open” is vacuously true, since it begins with “for any $x \in \emptyset$...”
To see that $\mathbb{R}$ is open, we must show that for any $x \in \mathbb{R}$ there exists $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subset \mathbb{R}$.
Of course any ϵ will do here, e.g. $\epsilon = 1$.
- (b) If $\bigcup_{i \in I} U_i = \emptyset$ then it is empty by part (a). Otherwise, suppose $x \in \bigcup_{i \in I} U_i$, i.e. $x \in U_i$ for some $i \in I$. Since U_i is open, there exists $\epsilon > 0$ such that

$$(x - \epsilon, x + \epsilon) \subset U_i \subset \bigcup_{i \in I} U_i,$$

as needed.

- (c) Again, if $U_1 \cap \dots \cap U_n = \emptyset$ then it is empty by (a). If not, then we can choose $x \in U_1 \cap \dots \cap U_n$, which means that $x \in U_i$ for all $1 \leq i \leq n$. For any $1 \leq i \leq n$, since U_i is open there exists $\epsilon_i > 0$ such that $(x - \epsilon_i, x + \epsilon_i) \subset U_i$. Let $\epsilon := \min(\epsilon_1, \dots, \epsilon_n)$. For any $1 \leq i \leq n$, we have $\epsilon \leq \epsilon_i$ and hence

$$(x - \epsilon, x + \epsilon) \subset (x - \epsilon_i, x + \epsilon_i) \subset U_i.$$

It follows that $(x - \epsilon, x + \epsilon) \subset U_1 \cup \dots \cup U_n$ as needed.

- (d) We claim that

$$\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

Clearly we have

$$\{0\} \subset \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right),$$

since $-\frac{1}{n} < 0 < \frac{1}{n}$ for any $n \in \mathbb{N}$. For the other inclusion, suppose that $x \in \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right)$, i.e. $x \in \left(-\frac{1}{n}, \frac{1}{n}\right)$ for any $n \in \mathbb{N}$. If we assume for the sake of contradiction that $x \neq 0$, then by the Archimedean property there exists $n \in \mathbb{N}$ such that either $\frac{1}{n} < x$ (if $x > 0$) or $-\frac{1}{n} > x$ (if $x < 0$). This means that $x \notin \left(-\frac{1}{n}, \frac{1}{n}\right)$, a contradiction. This shows that

$$\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \subset \{0\}.$$

2. Suppose for the sake of contradiction that $\sup A$ exists but $\sup A \notin A$, i.e. $\sup A \in \mathbb{R} \setminus A$. Since A is closed, i.e. $\mathbb{R} \setminus A$ is open, there exists $\epsilon > 0$ such that $(\sup A - \epsilon, \sup A + \epsilon) \subset \mathbb{R} \setminus A$, or equivalently $(\sup A - \epsilon, \sup A + \epsilon) \cap A = \emptyset$.

We claim that $\sup A - \epsilon$ is an upper bound for A . If $x \in A$, then certainly $x \leq \sup A$, since $\sup A$ is an upper bound for A . If $x > \sup A - \epsilon$, then we therefore have

$$\sup A - \epsilon < x \leq \sup A < \sup A + \epsilon,$$

i.e. $x \in (\sup A - \epsilon, \sup A + \epsilon)$. But this contradicts $(\sup A - \epsilon, \sup A + \epsilon) \cap A = \emptyset$, so we must have $x \leq \sup A - \epsilon$ as claimed.

But now $\sup A - \epsilon$ is an upper bound for A with the property that $\sup A - \epsilon < \sup A$, contradicting the fact that $\sup A$ is the *least* upper bound for A . We conclude that $\sup A \in A$.

3. (a) We claim that $\mathbb{Z}$ is closed. Namely, its complement is $\bigcup_{n \in \mathbb{Z}} (n, n+1)$, which is a union of open intervals and hence open by 1(b).

On the other hand, by the Archimedean property $\mathbb{Z}$ is not bounded. So the Heine-Borel theorem implies that $\mathbb{Z}$ is not compact.

- (b) First we show that $A := \{\frac{1}{n} \mid n \in \mathbb{N}\}$ is not closed. Namely, we have $0 \notin A$, and for any $\epsilon > 0$ we have $(-\epsilon, \epsilon) \cap A \neq \emptyset$ because the Archimedean property guarantees that there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$. Thus $\mathbb{R} \setminus A$ is not open, i.e. A is not closed.

Alternatively, one can use the analogue of 2 that says that if A is closed and $\inf A$ exists, then $\inf A \in A$ (either prove it in the same way as 2, or use the fact that $\inf A = -\sup(-A)$ to deduce it directly). Since $\inf A = 0 \notin A$ in this case by exercise 2(a) on Problem Set 3, it follows that A is not closed.

By the Heine-Borel theorem, since A is not closed, it is not compact.

- (c) To see that $A := \{\frac{1}{n} \mid n \in \mathbb{N}\} \cup \{0\}$ is closed, observe that

$$\mathbb{R} \setminus A = (-\infty, 0) \cup \left(\bigcup_{n \in \mathbb{N}} \left(\frac{1}{n+1}, \frac{1}{n} \right) \right) \cup (1, \infty)$$

is a union of open intervals and hence open.

Since $A \subset [0, 1]$ it is bounded, hence compact by the Heine-Borel theorem.

4. This is a difficult problem. If you're having trouble visualizing what this decomposition looks like in general, check out the formula for $\mathbb{R} \setminus A$ in 3(c) for a fairly complicated example.

We'll use the following lemma: if $A, B \subset \mathbb{R}$ are intervals and $A \cap B \neq \emptyset$, then $A \cup B$ is an interval. Let $x, y \in A \cup B$ be such that $x < y$, so we must show that $[x, y] \subset A \cup B$. If $x, y \in A$ then $[x, y] \subset A$ because A is an interval, and if $x, y \in B$ then $[x, y] \subset B$ because B is an interval. So we can assume without loss of generality that $x \in A \setminus B$ and $y \in B \setminus A$. Since $A \cap B \neq \emptyset$, there exists $z \in A \cap B$. If $z \leq x$, then since $y, z \in B$ and B is an interval we have $[z, y] \subset B$, but then $x \in [z, y] \subset B$, a contradiction. Similarly, if $z \geq y$ then since $x, z \in A$ and A is an interval, we have $[x, z] \subset A$, but then $y \in [x, z] \subset A$, contrary to assumption. So we have $x < z < y$. Given $w \in \mathbb{R}$ satisfying $x < w < y$, we need to show that $w \in A \cup B$. If $w = z$, we are done because $z \in A \cap B \subset A \cup B$. If $x < w < z$, then since $x, z \in A$ and A is an interval, we have $w \in [x, z] \subset A \subset A \cup B$. Similarly, if $z < w < y$, then $w \in [z, y] \subset B \subset A \cup B$. This covers all possible cases, so we obtain $w \in A \cup B$ as desired.

Next, we introduce some convenient terminology. Call $V \subset \mathbb{R}$ an *open subinterval* of U if V is an open interval and $V \subset U$. We say that an open subinterval $V \subset U$ is *maximal* if, for any open subinterval $W \subset U$ such that $V \subset W$, we have $V = W$.

Let I denote the set of maximal open subintervals of U . Since $V \subset U$ for any $V \in I$, we have

$$\bigcup_{V \in I} V \subset U.$$

To see the other inclusion, we must show that for any $x \in U$, there exists $V \in I$ such that $x \in V$. Let J_x denote the set of all open subintervals $W \subset U$ such that $x \in W$, and put

$$V_x := \bigcup_{W \in J_x} W.$$

Clearly $V_x \subset U$. We claim that $x \in V_x$ and that $V_x \in I$, i.e. V_x is a maximal open subinterval of U .

To see that $x \in V_x$, it suffices to show that $J_x \neq \emptyset$, i.e. there exists an open subinterval $W \subset U$ such that $x \in W$, since then we have $x \in W \subset V_x$. Indeed, since U is open there exists $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subset U$, and we can simply put $W := (x - \epsilon, x + \epsilon)$.

As for proving $V_x \in I$, we must first show that V_x is an interval. Given $y, z \in V_x$, we need to show that $[y, z] \subset V_x$. There exist open subintervals $W, W' \subset U$ such that $x \in W$, $x \in W'$, $y \in W$, and $z \in W'$. In particular $x \in W \cap W'$, so $W \cap W' \neq \emptyset$ and the lemma implies that $W \cup W'$ is an interval. But

then $y, z \in W \cup W'$ implies that $[y, z] \subset W \cup W' \subset V_x$, where the last inclusion holds by the definition of V_x .

To finish proving that $V_x \in I$, note that V_x is open by 1(b), since it is a union of open intervals. It remains to show that V_x is maximal among open subintervals of U . If $W \subset U$ is an open subinterval such that $V_x \subset W$, we must show that $V_x = W$, for which it suffices to prove that $W \subset V_x$. Note that $x \in V_x \subset W$, so W is an open subinterval of U containing x , i.e. $W \in J_x$. Thus, by the definition of V_x we have $W \subset V_x$.

This completes the proof that

$$U = \bigcup_{V \in I} V.$$

Next, we prove that for any $V, V' \in I$ such that $V \neq V'$, we have $V \cap V' = \emptyset$. Assume that $V \cap V' \neq \emptyset$ for the sake of contradiction. Then the lemma implies that $V \cup V'$ is an interval. It is open by 1(b), so $V \cup V'$ is an open subinterval of U . Since V and V' are maximal open subintervals, both of the inclusions $V \subset V \cup V'$, $V' \subset V \cup V'$ are equalities, i.e. we have $V = V \cup V'$ and $V' = V \cup V'$. This implies that $V = V'$, contrary to assumption.

Finally, it remains to show that I is finite or countably infinite. We will produce an injection $I \rightarrow \mathbb{Q}$, which by a previously given hint will imply that I is finite or countably infinite. Note that if $\emptyset \in I$, then $I = \{\emptyset\}$ and $U = \emptyset$. So we assume that $\emptyset \notin I$, meaning any $V \in I$ satisfies $V \neq \emptyset$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, for any $V \in I$ we can choose some $a_V \in \mathbb{Q} \cap V$. We claim that the map $I \rightarrow \mathbb{Q}$ given by $V \mapsto a_V$ is injective. Namely, if $a_V = a_{V'}$ for some $V, V' \in I$, then we have $a_V \in V \cap V'$ and hence $V \cap V' \neq \emptyset$. It follows that $V = V'$, since we have already shown that elements of I are pairwise disjoint.